

PROLOG PROGRAM FOR FRUITS AND THEIR COLOURS USING BACKTRACKING

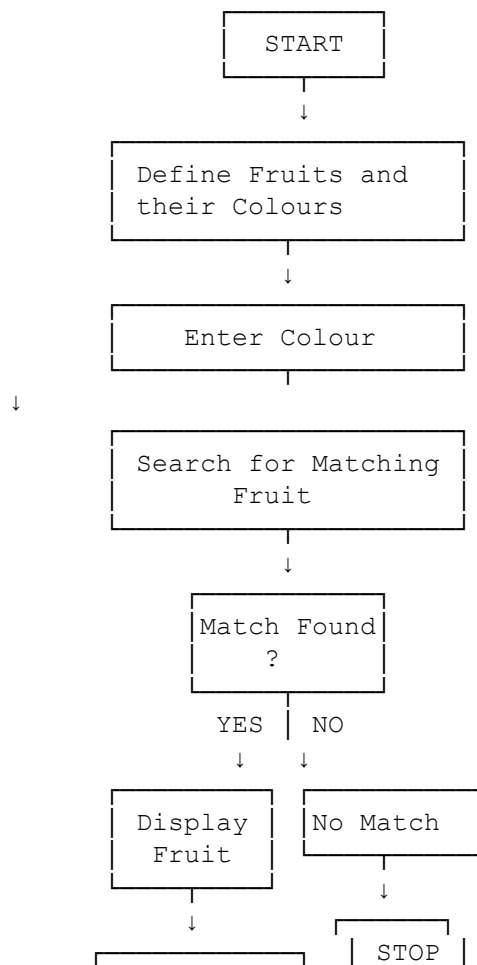
Aim

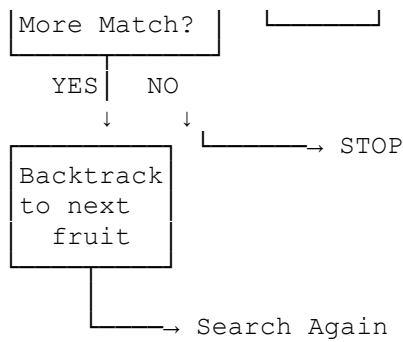
To write a Prolog program to identify fruits and their corresponding colours using backtracking.

Algorithm

1. Start.
2. Define facts for different fruits and their colours.
3. Create a predicate to find a fruit based on its colour.
4. Enter a colour as input.
5. Prolog searches for a fruit matching the given colour.
6. If multiple fruits match, use ; (**semicolon**) for backtracking.
7. Display all matching fruits.
8. Stop.

Flowchart





Prolog Program

```

% Fruit and Colour using Backtracking

fruit(apple, red).
fruit(banana, yellow).
fruit(orange, orange).
fruit(grapes, green).
fruit(mango, yellow).

find_fruit(Color, Fruit) :-
    fruit(Fruit, Color).
  
```

Output

```

?- find_fruit(yellow, Fruit).
Fruit = banana .

?- find_fruit(red, Fruit).
Fruit = apple.

?- 
  
```

Explanation

The query searches for fruits having the colour **yellow**. Prolog first finds **banana**. When `.` is pressed, Prolog **backtracks** and finds the next matching fruit, **mango**.